



Figure 1: SimpleCV dependencies

Installation

SimpleCV can be installed on all major operating systems. As it makes the computer vision code simple by providing a wrapper over existing libraries, it has many dependencies. Hence, during installation, all those dependencies need to be taken care of for making SimpleCV ready to run. The process of installation varies in different operating systems. The dependencies for SimpleCV are shown in Figure 1.

If you prefer to install SimpleCV in Ubuntu, use the following commands:

```
sudo apt-get install ipython python-opencv python-scipy
python-numpy python-pygame python-setuptools python-pip
sudo pip install https://github.com/ingenuitas/SimpleCV/
zipball/master
```

Once the installation has been successfully completed, run *simplecv* from the console.

For Windows operating systems, there are two different ways of installing SimpleCV. One method is to download the Windows Superpack from <http://www.simplecv.org/download/>. This super pack includes all the dependencies required for SimpleCV installation.

However, if your system already has some of the dependencies and you want to keep them as they are, then individual components can be downloaded separately, as shown below:

1. Download and install Python 2.7 using <http://www.python.org/ftp/python/2.7.3/python-2.7.3.msi>
2. The Python set-up tools for Windows can be downloaded from <http://pypi.python.org/packages/2.7/s/setuptools/setuptools-0.6c11.win32-py2.7.exe>
3. Installation of the SciPy super pack can be carried out using <http://sourceforge.net/projects/scipy/files/scipy/0.9.0rc5/scipy-0.9.0rc5-win32-superpack-python2.7.exe/download>
4. Installation of the NumPy super pack can be done by using <http://sourceforge.net/projects/numpy/files/NumPy/1.6.2/numpy-1.6.2-win32-superpack-python2.7.exe/download>
5. Installation of Pygame for Windows can be done using <http://pygame.org/ftp/pygame-1.9.1.win32-py2.7.msi>
6. Installation and configuration of OpenCV can be done by using <http://sourceforge.net/projects/opencvlibrary/>

<files/opencv-win/2.3.1/OpenCV-2.3.1-win-superpack.exe/download>

After the downloading has been completed, execute the .exe file and extract to the folder C:\OpenCV2.3\]. It can be extracted to any other folder too, but the environment paths are to be set accordingly.

After the installation is over, set the 'PATH' variables accordingly, as shown below:

```
SETX PATH C:/Python27/;C:/Python27/Scripts/;C:/OpenCV2.3/
opencv/build/x86/vc10/bin/;%PATH%
SETX PYTHONPATH C:/OpenCV2.3/opencv/build/
python/2.7/;%PYTHONPATH%
```

To update the 'PATH' variables, it is required that you close the current command prompt and reopen it. Then execute the following commands:

```
easy_installpyreadline
easy_install PIL
easy_installcymthon
easy_install pip
pip install ipython
pip install https://github.com/ingenuitas/SimpleCV/
zipball/1.3
```

After the successful installation by executing the above commands, type 'simplecv' from the command prompt at the installation folder, to open the *simplecv* shell. Otherwise, you may click on the SimpleCV icon on the desktop.

SimpleCV shell

The SimpleCV shell allows us to execute the commands and get the results in an interactive manner. Some of the shell interaction commands are as listed in Table 1.

Command	Description
<i>Exit</i>	Enables users to exit the SimpleCV shell. The same thing shall be achieved by pressing <i>Ctrl + D</i> .
<i>tutorial</i>	Allows users to start the SimpleCV interactive tutorial
<i>clear</i>	Performs the cleaning of the SimpleCV shell
<i>example</i>	Provides a collection of examples that we can run
<i>Forums</i>	Launches a Web browser for the help forums
<i>walkthrough</i>	Launches a Web browser with a walk-through

Table 1: SimpleCV shell interaction commands

The SimpleCV console is shown in Figure 2.

SimpleCV incorporates a detailed and elegant help system. The help on any of the components/libraries can be retrieved by using the *Help* command. An example of a command and its output is listed below: